Blessed G/Michael catholic secondary school
first semester Biology final examination for grade 12th

January 2018

Time allowed 1hour

Name --Hanna Ayalew-- section ---------- roll number ---------

I. Choose the best answer among the given alternatives.

A 1. The biochemical cycle of which of the following elements is based on changes in solubility rather than redox chemistry ?

A .Phosphorous ✓ B .Nitrogen × C .Sulphur × D .Carbon ×

C 2. Which of the following is the group of archae that can use CO_2 as their final electron acceptor during anaerobic respiration ? $CH_4 \rightarrow CO_2$

A .Metanotrophs × B .Methanogens C .Methylotrophs × D .An oxygenic photo synthesizers

B 3. During the light dependent reactions which molecule loses an electron ?

A .A reaction center pigment molecule × C .NADPH ×

B .A light harvesting pigment molecule ✓ D .Glyceraldehyde three phosphate

C 4. Which of the following does not occur during cyclic photophosphorylation in cyanobacteria ?

A .Electron transport through an electron transport chain C .NADPH formation ×

B .ATP synthesis D .Photosystem I use

C 5. Which of the following are two products of the light dependent reactions ?

A .Glyceraldehyde 3 phosphate and ribulose 1-5 biphosphate C .NADPH& ATP ✓

B .Glucose &O_2 D .All

C 6. Which of the following is not a name for the cycle resulting in the conversion of a two acetyl ,one ATP ,and two CO_2,one FADH and three NADH molecules ?

A .Tri carboxylic acid cycle C .Calvin cycle

B .Citric acid cycle D .Kreb's cycle

D 7. The link reaction is ?

A .Produces CO_2 ✓ C .Produces NADH ✓

B .Connects glycolysis with the kerb' s cycle ✓ D .all

C 8. Decarboxylation of pyruvate takes place ?

A .Outer mitochondrial membrane C .Cytosol

B .Mitochondrial matrix D .Inner mitochondrial membrane

A 9. What is the name of the molecule generated in the final step of the citric acid cycle and also used in the first step ?

A .Oxaloacetate B .Acetyl coA C .NADH D. Pyruvate

D 10. What is chemiosmosis ?

A .The movement of protons from high concentration to low concentration gradient through ATP synthase ✓

B .The movement of electrons through a series of electron carriers ×

C. The movement of protons from the inter membrane space to the matrix of mitochondria

D .A& C

D 11. In the kreb's cycle ?

A .ATP synthesis by oxidative phosphorylation takes place ×

B .ATP synthesis by substrate level phosphorylation takes place ✓

C .Decarboxylation reaction takes place

D .B & C

A 12. .In resting muscles most ATP transfers its energy ?

A .Creatine phosphate B .Lactic acid C .Phosphenol pyruvate D .all

B 13. Coenzymes that are derived from pantothenic acid is ?

A .Coenzyme Q B .Coenzyme A . C .NAD D .FAD

A 14. Antenna complex is ?

A .An array of proteins (enzymes and electron carriers) ,light harvesting complex embedded in the stroma of the chloroplast

B. An array of proteins (enzymes and electron carriers) ,light harvesting complex embedded in the thylakoid membrane of the chloroplast

C .Includes enzymes of dark reaction

D :All

C 15. Which of the following is true about C_4 plants ?

A .Fix atmospheric CO_2 by an enzyme rubisco during C_4 metabolism

B .Produces phosphoglycerate as the first stable product

C. Produces oxaloacetate as the first stable product

D .A& B

A 16. C3 plants are photo synthetically efficient in conditions of ?

A .High CO_2 concentration C .High light intensity

B .low CO_2 concentration D .High temperature

A 17. In photorespiration?

A .Low CO_2 concentration causes an enzyme phosphenol pyruvate to produce less phosphoglycerate than usual

B .Low CO_2 concentration causes an enzyme rubisco to produce less phosphoglycerate than usual

C .High CO_2 concentration causes an enzyme rubisco to produce less phosphoglycerate than usual

D. None

D 18. Which of the following is true about cyclic photophosphorylation ?

A .Only ps I is involved C .No need of reducing NADP

B .Electrons originates from psI and eventually returns to Ps I D .all

A 19. Which of the following is common to both light dependent and light independent reactions of photosynthesis ?

A .Redox reaction C .Synthesis of ATP

B .Photolysis of water D .Reduction of phosphoglycerate into trios phosphate

B 20. In C4 plants C4metabolissm occurs in ?

A .The chloroplasts of bundle sheath cells C .The stroma of bundle sheath cells

B .The chloroplasts of mesophyll cells D .In thylakoid membrane

B 21. Which enzyme is exclusively found in mesophyll cells of C_4 plants and is responsible the initial fixation of CO_2 ?

A .RUBISCO C .Pyruvate orthophosphate d kinase

B .PEP carboxylase D. NADP reductase

22. The calvin cycle is often referred to as the dark reaction but in reality it takes place most often in periods of sunlight this is because /

A .The enzymes of the calvin cycle are directly activated by photons

B .The reactions of the calvin cycle requires the continuous supply of ATP and NADPH from the light dependent reactions

C .CO_2 uptake by stomata only occurs during night

D The cycle produces oxygen which requires light

23. The primary advantage of the C4pathway over the C_3pathway under high light intensity ,high temperatur and arid conditions is the ability to ?

A .Directly uses sunlight for CO_2 fixation in mesophyll cells

B .Minimizes photorespiration by concentrating CO_2 around RUBISCO in the bundle sheath cells

C .Produces ATP and NADPH more efficiently in the light dependent reactions

D .Fix CO_2 exclusively at dark

24. CAM plants open their stomata ?

A .During day

B .At night

C .Only when it rains

D .Never

25. Which of the following are directly associated with psI ?

A .Harvesting of light energy by ATP

B.P-680 reaction canter molecule

C. Recieving electrons from plastocyanin

D .Passing electrons to plastoquinone

26. In chloroplast chemiosmosis trans locates protons to the ?

A .The stroma to the thylakoid space

B .The stroma to psII

C .ATP synthase to NADP reductase

D .The matrix to the stroma

27. Which pigment captures light energy in ps II ?

A .Chlorophyll a (p-680)

B .Chlorophyll a (p700)

C .Carotenoids

D .Chlorophyll b

28. Where does linear photophosphorylation takes place ?

A .Thylakoid of the chloroplast

B .Stroma lamella

C .Stroma of chloroplast

D .None

29. In cyclic photophosphorylation the electrons that returns to ps I after electron transport chain is transferred to ?

A .Plastoquinone B .NADP C .Cytochrome complex D .Oxygen

30. In chromatography ,which pigment travels the farthest (highest Rf value ?

A .Chlorophyll b B .Chlorophyll a C .Carotene D .Xanthophyll

31. Which element is at the center of porphrin ring in chlorophyll ?

A. Manganese B . Magnesium C .Iron D .Calcium

32. What does the action spectrum represents ?

A .The rate of photosynthesis at different wavelengths of light

B .The light entitled by photosynthetic organisms

C .The wavelengths of light absorbed by chlorophyll a

D .The amount of light energy absorbed by all pigments

33. What is the primary reason plants use multiple pigments (chlorophyll and carotenoids)

A .To capture a wide range of light wavelengths ,maximizing energy capture.

B, To reflect green light more efficiently

C .To produce more oxygen

D .To allow of photosynthesis in dark

34. Which process occurs in both aerobic and anaerobic respiration ?

A .Citric acid cycle

B. Fermentation

C .Pyruvate dehydrogenase complex

D .Glycolysis

35. Which of these is not true of fermentation ?
A .NADH donates electrons to the ETC
B .Follows glycolysis
C .Starts with glucose
D .Carried out by yeast

36. Glycolysis is thought to be one of the most ancient metabolic processes ,which statement supports this idea ?
A .Glycolysis is used by all cells
B .Glycolysis neither uses nor needs oxygen
C .Cells were performing glycolysis long before oxygen was present in earth's atmosphere
D .All

37. During photorespiration ,where does decarboxylation of glycine to form serine and release of carbon dioxide takes place ?
A. Peroxisomes B .Mitochondria C .Chloroplast D .Cytosol

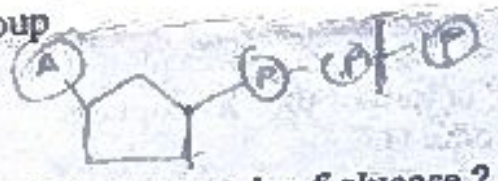
38. What is the stable product formed in the photorespiration cycle in the chloroplast ?
A .Oxaloacetate B .Phosphoglycerate C .Phosphoglycolate D .G3P

39. Clusters of light gathering antenna pigment in a photosystem is ?
A Passes energy to the reaction centre
B .Do not absorb photons
C .Absorbs electrons
D .Break down H2O

40. Photophosphorylation differs from oxidative phosphorylation in that ?
A The final electron acceptor is NADP but not oxygen
B .It involves ETC
C .Energy stored in the form of a proton concentration difference
D. Regeneration of ATP is driven by a flow of protons through ATP synthase

41. Which enzyme is responsible for the massive synthesis of ATP during oxidative phosphorylation in cristae ?
A .ATPase B .ATP synthase C .Kinase D. Transaminase

42. Which part of the ATP molecule is responsible for the high energy and readily releasable energy
A. The bond between the first and the second group
B .The bond between second and third group
C .Adenine base
D .Ribose sugar



43. How many turns of the calvin cycle is required to synthesize one mole of glucose ?
A .1 B.2 C .6 D .3

44. How many moles of ATP is synthesized from 3 moles of pyruvic acid during the TCA cycle ?
A .2 B .3 C.6 D.4

45. What is the primary function of ETC ?
A .To create a proton gradient (proton motive force) to drive ATP synthase
B .To produce pyruvate
C .To directly generate ATP via substrate level phosphorylation
D .To break down glucose into CO_2

46. Cyanide inhibits ETC by binding to which complex ?
A .Complex II C ,Complex IV (Cytochrome c oxidase)
B .Complex III D .Complex I

47. Which one of the following electron donors in ETC donates its electrons first at ubiquinone ?
A .NADH B .FADH C .both D. none

48. Which of the following vitamins is required for the synthesizing a cofactor required for converting succinate to fumarate ?
A. Thiamine B .Riboflavin C .Panthotenicacid D .Niacin

49. Which of the following enzymes is involved in the formation of GIP by phosphorylating GDP ?
A .Succinate thiokinase C .Succinate dehydrogenase
B .Citrate synthase D .ATPase

50. A73 year old male is brought to emergency with acute myocardial infarction ,blood chemistry reveals lactic acidosis .How much energy yield per mole of glucose is expected in such a condition expected ?

A .2 B .6 C 8 D .10

51. Which experimental observation provides evidence that oxygen is produced during photosynthesis ?

A .The production of starch in plant leaves

B .The disappearance of CO_2 in the presence of light

C .The accumulation of bubbles from a submerged aquatic plant under light

D .A decrease in the surrounding temperature

52. When presented with a mixture of glucose and fructose, which sugar will yeast preferentially use first?

A .Fructose B .Glucose C .They use both at the same rate D .Neither

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Answer Sheet

37/50

Name Hanna Ayalew Section 12TC No 24

1. A	12. A	23. B	33. A	43. C
2. BC	13. B	24. B	34. D	44. B
3. BA	14. AB	25. C	35. A	45. A
4. C	15. C	26. DA	36. D	46. AC
5. C	16. A	27. A	37. B	47. CB
6. C	17. AB	28. BA	38. BC	48. B
7. D	18. D	29. CA	39. DA	49. C.A
8. CB	19. A	30. DC	40. A	50. DA
9. A	20. B	31. B	41. B	51. C
10. D	21. B	32. A	42. B	52. B
11. D	22. B			